AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 11

Lab Title Graphical User Interface

Student Name: Ali Abbas Khan

Reg. No: 230661

Objective: _____

_____

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 11

Title: GUI

Name: Ali Abbas Khan

Roll Number: 230661

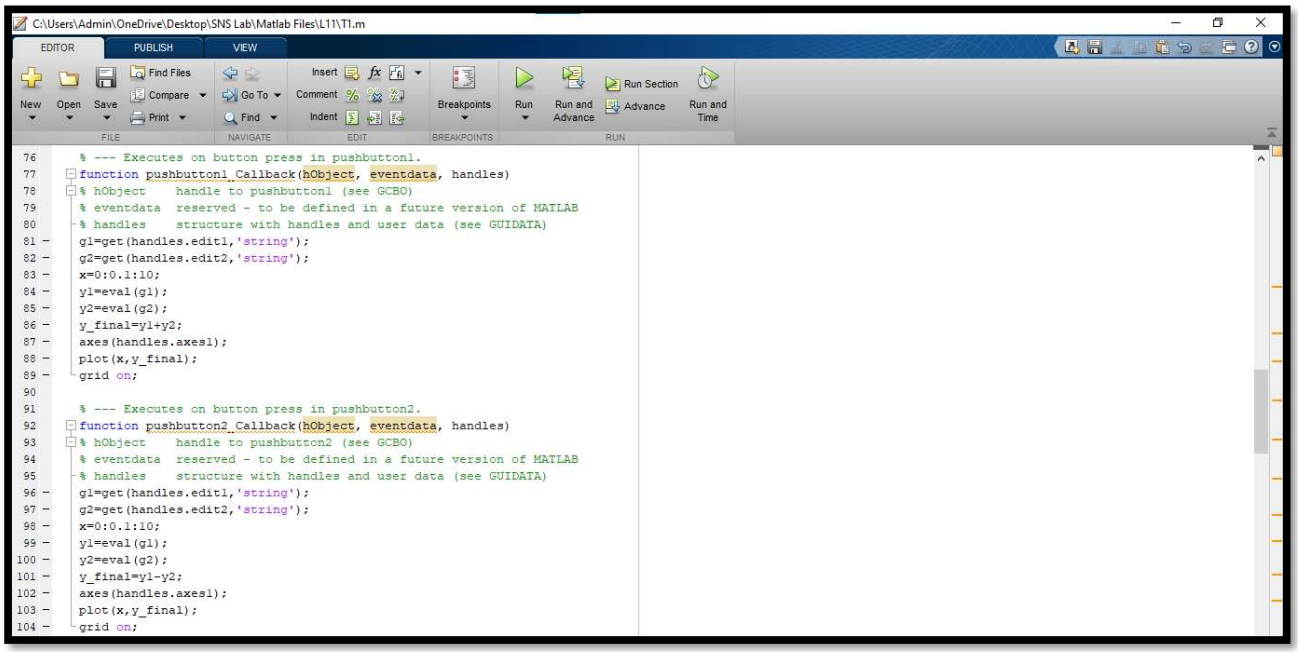
Class: BCE-5B

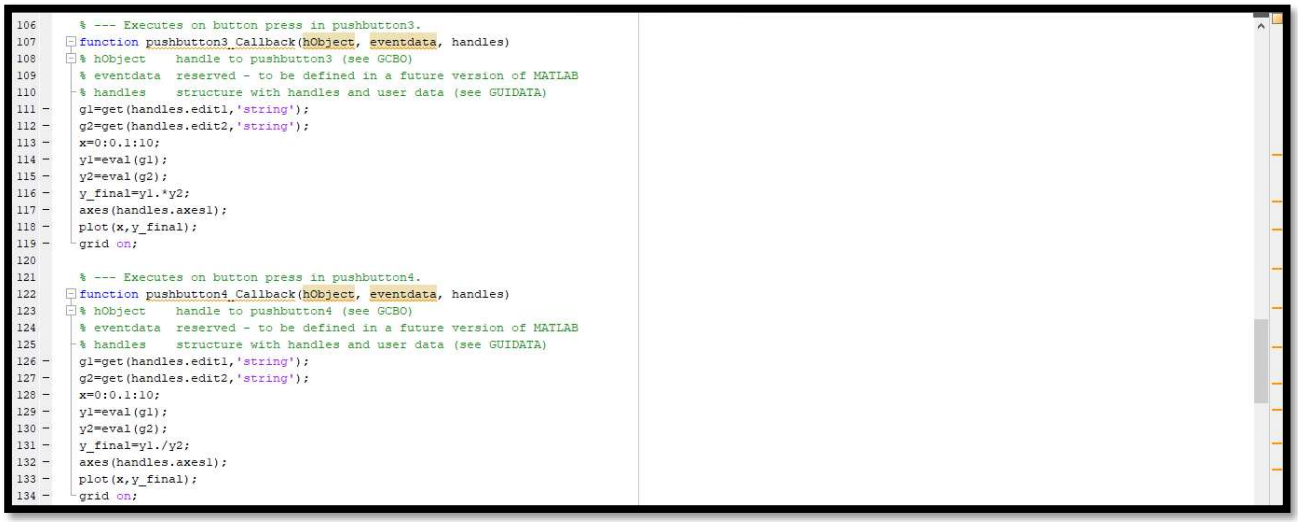
Submitted To: Engr. Maryam
Date**:** December 10, 2025

Lab Task:

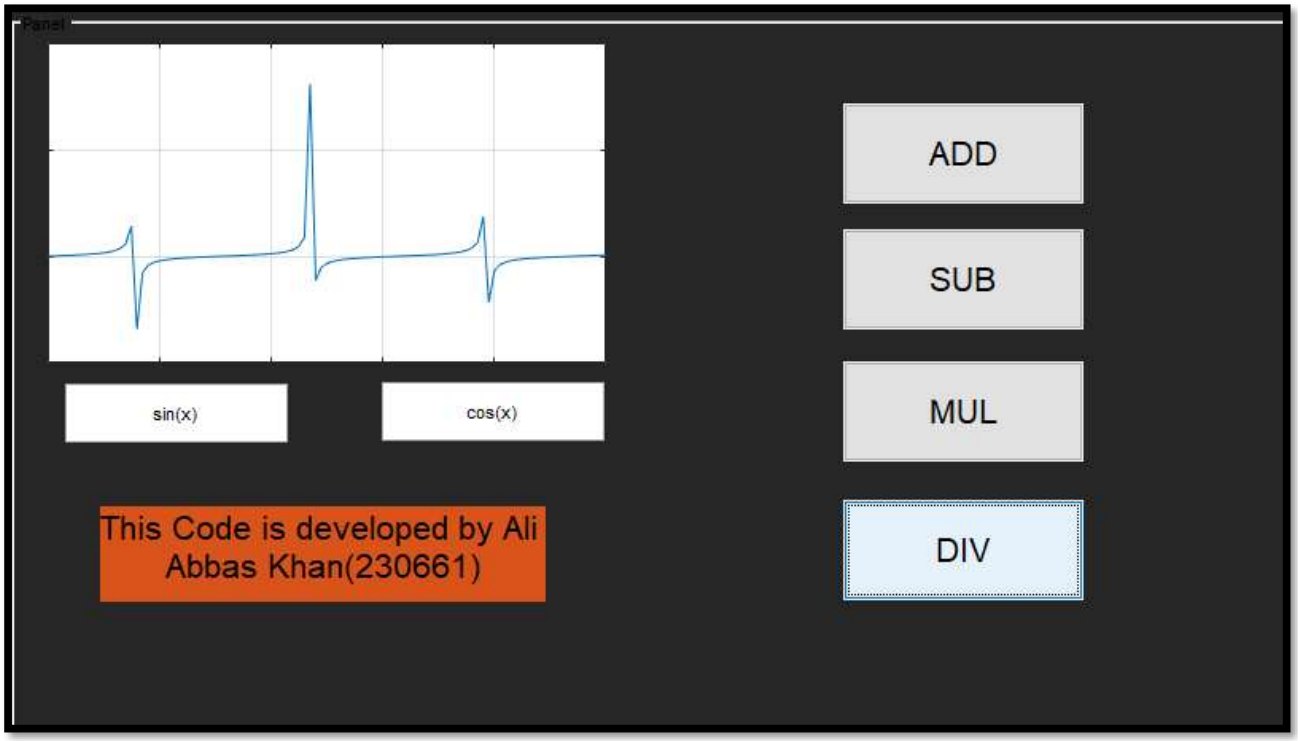
1. Design a simple graph using MATLAB GUI that can add, subtract, multiply and divide two cosine and sine signal and displays the result.Make Sure this should be colorful and presentable.

   Code:





Output:



Consider the given function $x(t)$ where:

$$x(t) = t^2 \ for -1 \leq t \leq 1$$

Calculate the Fourier series coefficients (a_k , b_k and a_0) using MATLAB. Use these coefficients to approximate $x(t)$.

Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L11\T2.m

T2.m | T3.m | T1.m

```
1 -    syms t n
2 -    T0=2*pi;
3 -    w0=1;
4 -    n=-1:1;
5
6 -    x=t^2;
7
8 -    a0=(1/T0) * int(x,t,0,T0);
9
10 -   an=(2/T0) * int(x*cos(n*w0*t),t,0,T0);
11 -   bn=(2/T0) * int(x*sin(n*w0*t),t,0,T0);
12
13 -   x_new=a0+bn.*cos(n*t)+bn.*sin(n*t)
14 -   x_new=vpa(x_new,length(n))
```

Output:

```
>> T2

x_new =

[ (1911387046407553*pi^3)/4503599627370496 + (5734161139222659*pi^2*cos(t))/4503599627370496 - (5734161139222659*pi^2*sin(t))/4503599627370496, (1911387046407553*pi^3

x_new =

[ 12.6*cos(t) - 12.6*sin(t) + 13.2, 13.2, 13.2 - 12.6*sin(t) - 12.6*cos(t)]
```

x_new =

[(1911387046407553*pi^3)/4503599627370496 +
(5734161139222659*pi^2*cos(t))/4503599627370496 -
(5734161139222659*pi^2*sin(t))/4503599627370496,
(1911387046407553*pi^3)/4503599627370496,
(1911387046407553*pi^3)/4503599627370496 -
(5734161139222659*pi^2*cos(t))/4503599627370496 -
(5734161139222659*pi^2*sin(t))/4503599627370496]

x_new =

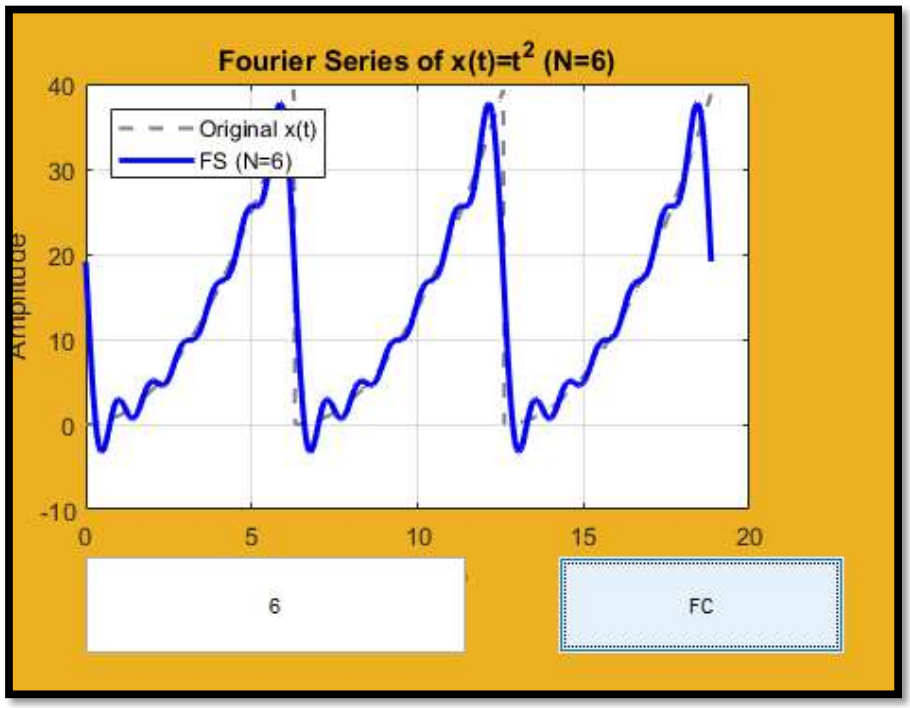
[12.6*cos(t) - 12.6*sin(t) + 13.2, 13.2, 13.2 - 12.6*sin(t) - 12.6*cos(t)]

Task 3:

```
100   function pushbutton1_Callback(hObject, eventdata, handles)
101   % hObject    handle to pushbutton1 (see GCBO)
102   % eventdata  reserved - to be defined in a future version of MATLAB
103   % handles    structure with handles and user data (see GUIDATA)
104 - N_terms_str = get(handles.edit1, 'String');
105 - N_terms = str2double(N_terms_str);
106 - if isnan(N_terms) || N_terms <= 0 || floor(N_terms) ~= N_terms
107 -     errordlg('Enter a positive integer for N.', 'Invalid Input');
108 -     return;
109 - end
110 - syms t n
111 - T0 = 2*pi;
112 - w0 = 1;
113 - x_orig = t^2;
114 - a0 = (1/T0) * int(x_orig, t, 0, T0);
115 - an_sym = (2/T0) * int(x_orig * cos(n*w0*t), t, 0, T0);
116 - bn_sym = (2/T0) * int(x_orig * sin(n*w0*t), t, 0, T0);
117 - x_approx_sym = a0;
118 - for k = 1:N_terms
119 -     an_k = subs(an_sym, n, k);
120 -     bn_k = subs(bn_sym, n, k);
121 -     x_approx_sym = x_approx_sym + an_k * cos(k*w0*t) + bn_k * sin(k*w0*t);
122 - end
123 - t_range = linspace(0, 3*T0, 500);
124 - x_approx_num = double(subs(x_approx_sym, t, t_range));
125 - x_orig_single_period = double(subs(x_orig, t, t_range(t_range <= T0)));
126 - x_orig_periodic = repmat(x_orig_single_period, 1, 3);
127 - x_orig_periodic = x_orig_periodic(1:length(t_range));
128 - axes(handles.axes1);
129 - cla;
130 - plot(t_range, x_orig_periodic, '--', 'Color', [0.5 0.5 0.5], 'LineWidth', 1.5, 'DisplayName', 'Original x(t)');
```

```
131 -    hold on;
132 -    plot(t_range, x_approx_num, 'b', 'LineWidth', 2, 'DisplayName', sprintf('FS (N=%d)', N_terms));
133 -    title(sprintf('Fourier Series of x(t)=t^2 (N=%d)', N_terms));
134 -    xlabel('Time (t)');
135 -    ylabel('Amplitude');
136 -    grid on;
137 -    legend('show', 'Location', 'NorthWest');
138 -    hold off;
```

Output:



Conclusion:

This lab successfully utilized MATLAB's symbolic capabilities within a GUIDE environment to calculate and visualize the Fourier series approximation of the function x(t) = t^2 over the interval [0, 2*pi]. By implementing the formulas for the DC component and the harmonic coefficients, the code iteratively constructed the series sum up to an arbitrary number of terms (N) defined by the user input. The resulting plot, displayed on the axes, clearly demonstrated how increasing the value of N leads to a more accurate approximation of the periodic function, visually illustrating the power of Fourier analysis in decomposing and reconstructing complex periodic signals.